

kernel

$$\begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & -2 & -1 & 0 \\ -1 & -2 & 16 & -2 & -1 \\ 0 & -1 & -2 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{bmatrix}$$

Edge detection

Aryan Watsp.

Max Hildreth

$\nabla^2 G \rightarrow$ Applied on image

$$\nabla^2 G (f(x, y))$$

Gaussian is applied $\Rightarrow$ smoothening noise reduction

Gaussian $\Rightarrow$ no abrupt transmission w.r.t. mem. filter transitions is smooth.

$\Rightarrow$ No ringing effect

$\nabla^2 G$ gaussian operations are linear

$$\nabla^2 G (f(x, y)) \quad 2^{\text{nd}} \text{ order derivative.}$$

$\rightarrow$ zero crossing pts can be taken as edges.

Zero crossing detection

Let P be a point
consider 3×3 neighbourhood centred at P
either at least one pair of opposite
pixels (top-bottom, left-right, diag-opp)
are of opposite sign for $\nabla^2 G$ output

Lanny Edge detector

Objective

- To minimize the detection error
 - True edges will not be missed
 - false detection of edges to be minimized
- Edges must be well localized.
 - detected edges must be as close as possible to the actual edge
- for every true edge point there should be one detected edge pixel.

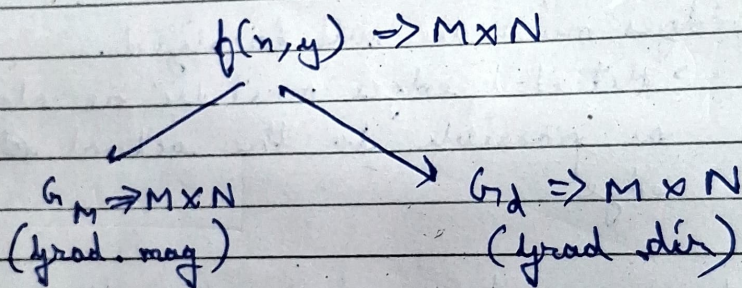
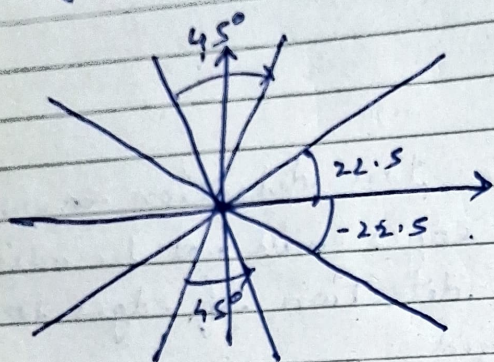
steps

- I. Gaussian smoothing (to reduce noise)
- II. Compute gradient magnitude & direction (use any kernel)
- III. Non maxima suppression (to make the edges thin)

grad. dir $\Rightarrow$ quantized into no. of few directions
 d_1, d_2, d_3, d_4

find the d_i closest to G_d

Canary edge detector



for every pixel at (n, y)

Let d_i be the quantized direction
 Consider, 3×3 neighbourhood around the pixel

if its mag $G_M(n, y)$ is greater than all the neighbours with direction d_i

$\rightarrow Q_M(n, y) = G_M(n, y)$ else $Q_M(n, y) = 0$
 matrix after non maxima suppression

IV Double thresholding to detect & link the edges

threshold is large $\Rightarrow$ false (-)ive
(miss the edge pixels)

threshold is small $\Rightarrow$ false (+)ive
(more pixels qualify to become edge)

double threshold T_H (high threshold)

Hysteresis T_L (low threshold)

if $g_N(n, y) > T_H$ $g_{NH}(n, y) = g_N(n, y)$ else
 $g_{NH}(n, y) = 0$

if $g_N(n, y) > T_L$ $g_{NL}(n, y) = g_N(n, y)$
else $g_{NL}(n, y) = 0$

$g_{NL}(n, y) = g_{NL}(n, y) - g_{NH}(n, y)$
(weak edge points)

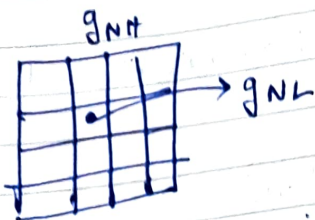
strong edges

perous lots of discontinuities
broken edges

look into g_{NL} into the neighbourhood of (n, y)
for each non zero pixel, $g_{NH}(n, y)$
{ if there exists any non zero $g_{NL}(n', y')$ then mark it }

for every marked pixel $g_{NL}(n, y)$
 $g_{NH}(n, y) = g_{NL}(n, y)$

$\hookrightarrow$ detected edges.



~~Start~~ further local checking

for every so far detected points, if there is discontinuity, look into its neighbourhood w.r.t certain properties.

Neighbourhood S

$$(n', y') \in S$$

$$|Mag(n, y) - Mag(n', y')| \leq M_{th}$$

$$|grad\ dir(n, y) - grad\ dir(n', y')| \leq d_{th}$$

Subail notes

for each point (x_i, y_i)

{

for each possible value of a_i

{

find $b_i = -a_i x_i + y_i$
increment $A[a_i][b_i]++$

}

}

OR

for each pt (x_i, y_i)

{

for each possible value of b_i

{

find $a_i = (y_i - b_i) / x_i$

$A[a_i][b_i]++$

}

}

Apply a threshold on $Acc[i][j]$

In the acc, ~~also~~ along with count, keep
 $x_{min}, x_{max}, y_{min}, y_{max}$

segmentation

Discont based
(edge line etc
detection)

~~Homogeneity~~ Homogeneity based
(region extraction)

Region extraction

$$R \Rightarrow R_1, R_2, \dots, R_n$$

$$\bigcup_i R_i = R$$

$R_i \Rightarrow$ connected

$R_i \cap R_j = \emptyset$ disjoint $i \neq j$

$\text{Prop}(R_i) = \text{True}$ homogeneous

$\text{Prop}(R_i \cup R_j) = \text{false}$ R_i, R_j are adjacent

~~Region growing~~

~~Region split~~

~~Region merge~~

~~split & merge~~

~~Region formation~~

Region extraction

pixel based
approach

feature/property
thresholding

simplest $\Rightarrow$ intensity

selection of

feature.

& threshold

Region based approach

Region
formation

region growing

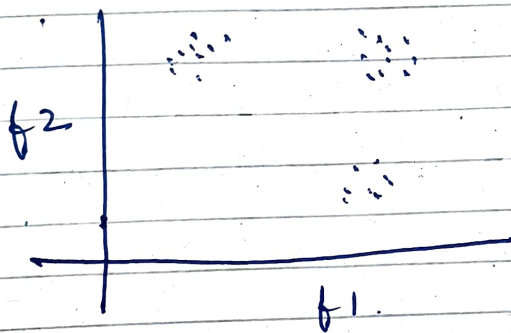
region split

region merge

split & merge

pixel based approach

- for every pixel calculate the feature
- study the feature distance in feature space.
- for different type of regions, different groups / clusters will be formed in the feature space.
- find out the discriminating boundary to isolate the clusters.



One dim. feature / property.

↓
dist. of feature can be approximated as histogram.

discrimination $f(x) \Rightarrow$ threshold.

2 types of regions

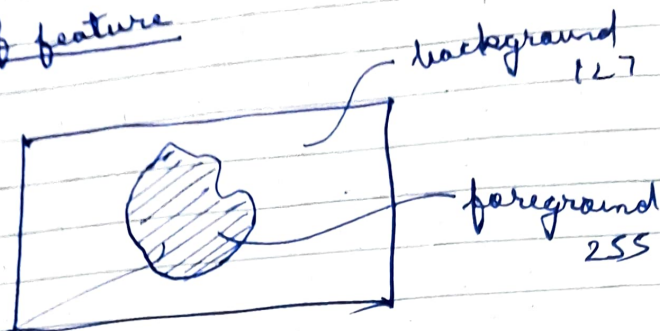
feature value, t is the threshold

$$f(x, y) > t \Rightarrow \text{old}(x, y) = 1$$

else 0

↓
features

choice of feature



intensity $\Rightarrow$ feature
threshold $\Rightarrow t$ ($> t$ object)



127 to 254

25% 255
50% 191
25% 127

uniformly distributed

$\rightarrow$ mean filter can be used
191 is then the threshold

if 0 25%
127 50%
255 25%

gradient filter can be used

selection of threshold (based on histogram)



2 types of regions

feature value is high for one region.

if known that $\Rightarrow$ p% of pixel belong to high featured region.

threshold t

feature histogram $\rightarrow h$

Image $\Rightarrow M \times N$

$$\sum_{i=t+1}^L h[i] = \frac{p * M * N}{100}$$

max feature value

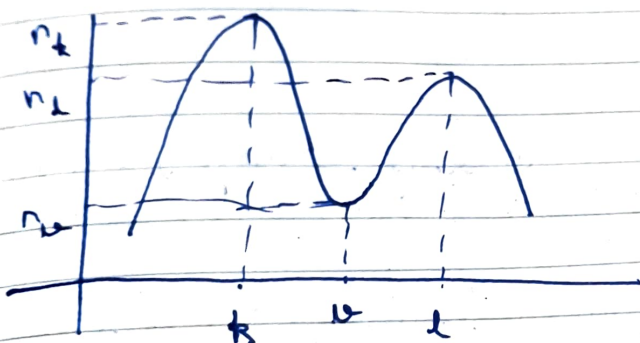
smallest value of t such that

$$\sum_{i=t+1}^L h(i) \geq p * \frac{MN}{100}$$

sum = 0;

for ($i=L$; $i > t$; $i--$)

{
 sum = sum + h[i]
 if (sum > $p * \frac{MN}{100}$) break;



2 types of regions

Bimodal $\Rightarrow$ modes denote frequently occurring property values

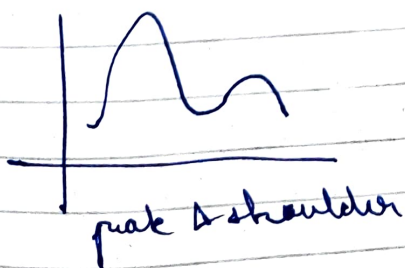
deep point/valley $\Rightarrow$ threshold

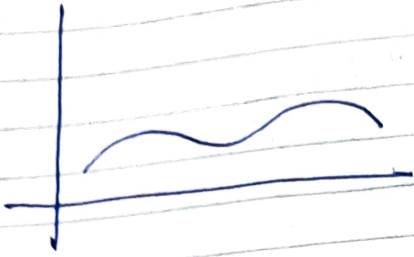
$$s_1 = \frac{|n_k - n_v|}{|v - k|}$$

$$s_2 = \frac{|n_l - n_v|}{|v - l|}$$

if $s_1 > th$ & $s_2 > th$ $\Rightarrow$ bimodal

if $s_1 > th$ but $s_2 \leq th$ or vice versa $\hookrightarrow$ peak & shoulder





flat $\Delta 1$ & $\Delta 2$ both are
 $\Phi < th$

Bimodality strength = $\Delta 1 * \Delta 2$

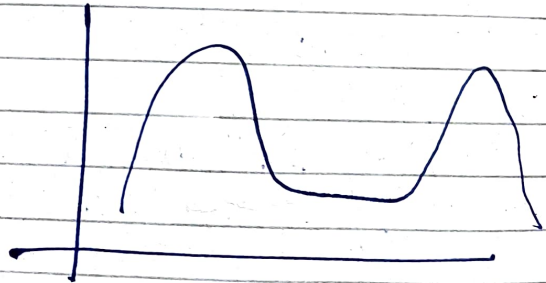
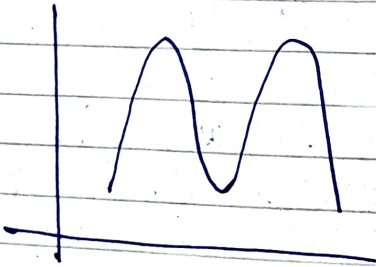
Bimodal

└ strongly
 └ weakly

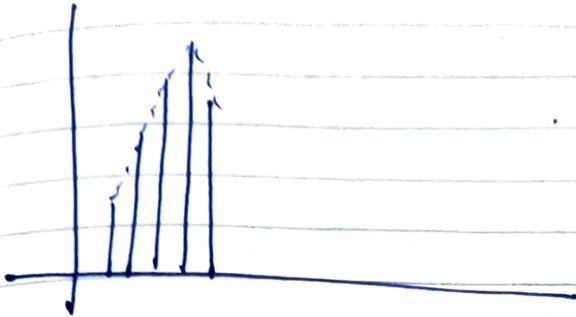
→ because of noise & blurring → deep valley may not be found

→ overlap in feature values between the regions

→ boundary of the regions may overlap rather than object interiors

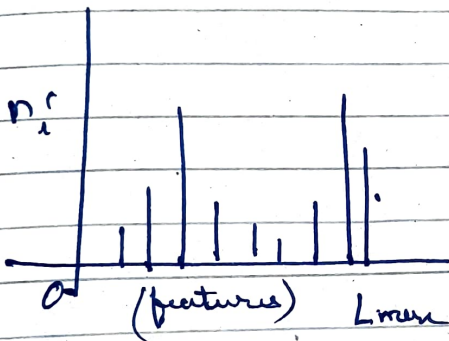


Histogram is ~~non~~ non-continuous $\Rightarrow$ discrete values
 $\Downarrow$
 gives rise to jaggedness



Finding valley is non trivial

valley detection



$\rightarrow$ Apply mean filter

$$\bar{n}_i = \frac{1}{2q+1} \sum_{j=i-q}^{i+q} n_j$$

$\rightarrow n_i = \bar{n}_i$

$\rightarrow$ find local maxima in a $2s+1$ window
 at $z_1, z_2, \dots, z_k$

$$n_{z_k} = \max \{ n_i \} \quad z_k - s \leq \quad \leq z_k + s$$

Region adjacency graph

- Each detected region is a node in the graph
- Regions which are adjacent (common boundary) corresponding nodes will be connected by edge

$R_i \& R_j$ are two connected nodes

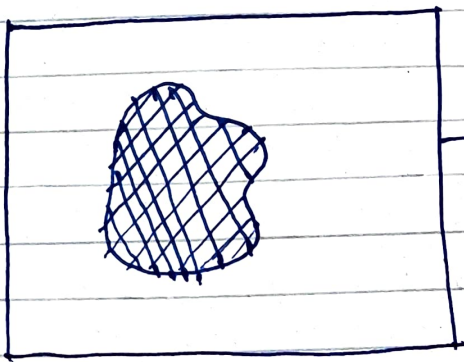
~~Shape~~

(NOTES MISSED)

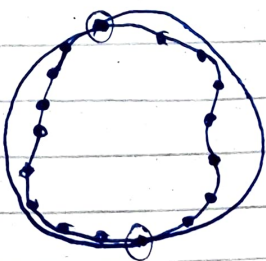
Shape descriptor

- Pre processing
- segmentation
- Components
 - (component labelling)
 - (individual component)

shape



smallest circle that encompasses the object



Find the most distant pair of points
↳ distance → diameter of circle.

circularity $\rightarrow$ object area $\frac{\pi d^2}{4}$

- $\rightarrow$ independent of size/scale
- $\rightarrow$ independent of translation
- $\rightarrow$ independent of rotation

object perimeter (no of boundary pixels)

$$\pi d$$

$(\bar{x}, \bar{y})$ center of gravity of the object

diameter $\Rightarrow$ mid pt center

$\{r_i\} \Rightarrow$ dist of boundary pixel from centre

median $\{r_i\}$ $d/2$

Convex hull

Smallest polygon encompassing the object such that line segment connecting any two object boundary pixels will completely remain inside the polygon.

Geometric moments

$$m_{i,j} = \sum_n \sum_y n^i y^j f(n,y)$$

moment of order i, j

$$m_{0,0} = \sum \sum f(n,y) \rightarrow \text{mass}$$

$$m_{1,0} = \sum \sum n f(n,y)$$

moment abt y axis

about x axis $m_{0,1} = \sum \sum y f(x,y)$

$$m_{1,1} = \sum \sum x y f(x,y)$$

$$m_{0,2} = \sum \sum y^2 f(x,y) \rightarrow \text{curtosis about } x \text{ axis}$$

$$m_{2,0} = \sum \sum x^2 f(x,y) \Rightarrow \text{about } y \text{ axis}$$

centralised moment (moment about centroid)
[~~indep~~. Independent of translation]

geometric moments

$$\bar{x}, \bar{y}$$

$$(\bar{x}, \bar{y}) = \left(\frac{m_{1,0}}{m_{0,0}}, \frac{m_{0,1}}{m_{0,0}} \right)$$

whole mass can be placed at centroid.

$$\bar{m}_{i,j} = \sum \sum (x - \bar{x})^i \cdot (y - \bar{y})^j f(x,y)$$

Independent of size

normalized central moment ~~$\bar{m}_{0,1} - m_0$~~

$$\cdot \mu_{i,j} = \frac{\bar{m}_{i,j}}{m_{0,0} \left(\frac{i+j}{2} + 1 \right)}$$

$$\bar{m}_{0,0} = m_{0,0}$$

$$\bar{m}_{0,1} = \sum \sum (y - \bar{y}) b(u, y)$$

$$= \sum \sum y b(u, y) - \bar{y} \sum \sum b(u, y)$$

$$= m_{0,1} - \frac{m_{0,1}}{m_{0,0}} \cdot \cancel{m_{0,0}}$$

Rotation Independent

→ eqns are formulated using 0/2/3 order normal central moment

↙ independent of size & rotation.

(SEE WHISP NOTES)

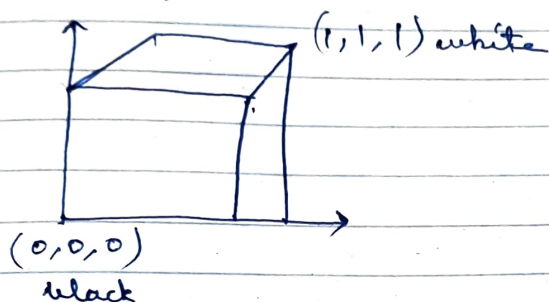
colour model



To describe colour in a formal way

RGB model

⇒ follows cartesian coordinate system where area of interest is a unit cube.



CMY model (cyan magenta yellow)

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

→ subtractive primaries

$\begin{matrix} C \\ M \\ Y \\ B \end{matrix} \left\{ \begin{array}{l} \rightarrow \text{used in real life} \\ \text{because } \text{CMY} \end{array} \right.$

→ black

YIQ model (for color transmission)

→ downward compatibility

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.30 & 0.59 & 0.11 \\ 0.60 & -0.32 & -0.28 \\ -0.52 & 0.21 & 0.31 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

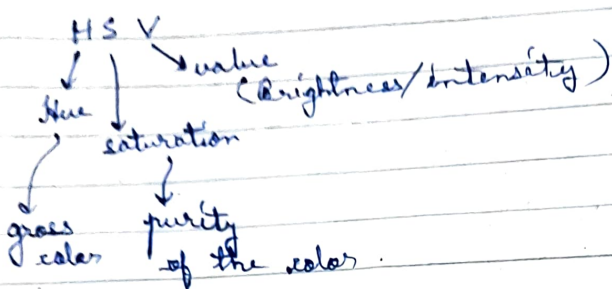
Intensity

Impulse

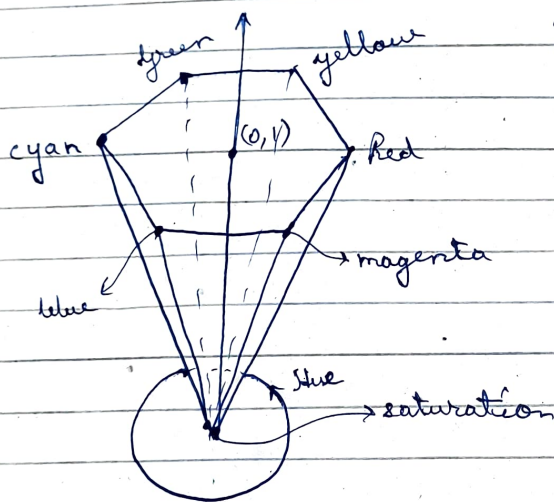
quadrature

Processing

↓
Segregating colour info from rest



Represents the color as an inverted hex ~~cone~~ cone



Tip is placed at the centre of the hue circle

0-360

sat $\Rightarrow [0, 1]$
value $\Rightarrow [0, 1]$

normalised R, G, B

$$\max = \max \{ R, G, B \}$$

$$\min = \min \{ R, G, B \}$$

if $R = \max$

$$h = \frac{G - B}{\max - \min}$$

if $G = \max$

$$h = 2 + \frac{B - R}{\max - \min}$$

if $B = \max$

$$h = 4 + \frac{B - G}{\max - \min}$$

if $h \geq 0$ then $H = h * 60$
 else $H = 360 + h * 60$

$S = \frac{\min}{\max}$ } if R, G, B are same
 $V = \max$ } $\Rightarrow$ the undefined

measure of grayness

colour feature

→ colour correlogram

	$\frac{R}{256}$	$\frac{G}{256}$	$\frac{B}{256}$

colour histogram

R/G/B



4 bin 4x4x4
 2 2 2